WORK PLAN		Version:
Project title:		Date: 14/9/2026
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Background (facts, current situation and organisational goals)

Avans University of Applied Sciences currently has an unfinished open-return wind tunnel available in the Aerospace Engineering laboratory. The main mechanical structure of the wind tunnel is already present, but the system is not yet fully operational and does not currently provide the measurement and data-acquisition functionality required for aerodynamic experiments.

The purpose of this project is to further develop the existing wind tunnel into a functional aerodynamic test facility that can be used for educational purposes. The system should allow students to perform practical ex periments on airfoils and compare physical measurements with theoretical aerodynamic models.
A major part of the project is the development and integration of an instrumentation and data-acquisition system. Static pressure measurements on an airfoil will be used to determine the pressure distribution over the airfoil. These measurements will subsequently be processed into a pressure coefficient $C_p$and plotted against the dimensionless chord position x/c.

The experimental results should be compared with theoretical results obtained using XFOIL. In this way, the wind tunnel will provide students with a physical test setup that connects aerodynamic theory, simulation, measurement technology and data processing.

Problem (description of the undesired situation)

-missing software to run the wind tunnel
-wrong sensor suite
-minor hardware problems (leaks, loose door)
-flow regulator (honeycomb structure) missing

Objective (description of the desired situation)

Complete the wind tunnel so it can be used for educational purposes.

Results and deliverables (what will be completed by the end of the project, including intermediate results)

A wind tunnel suitable for:
- -measuring Cp agains x/c diagrams.
- -The ability to work standalone
- -Plot the data live on a display
- -Record the data to a usb drive
- -Adjust the angle of attack (manually and elek
- -Set desired windspeed

Documentation for:
- -A user manual for operating and using the windtunnel
- -A database with research, software and electrical schematics
- -A motivation for project decisions
- -A global planning

-Test results for validating it's functioning

Scope exclusions (what is outside the project)
-airspeeds above mach 0.3 -high accuracy measurements - Achieving an exact correlation between the theoretical XFOIL simulations and the practical wind-tunnel measurements is outside the scope of this project. The project focuses on comparing both results and identifying and explaining possible deviations.

Risks	Likelihood	Impact	Mitigation measures
Airflow in the test section is insufficient or unstable	Mid	High	Measure the actual flow conditions, locate major losses/leaks and improve local flow conditioning where possible. Select a practical operating velocity rather than targeting the original maximum airflow.
Pressure sensors have insufficient range or resolution	Mid	High	Bench-test and calibrate the sensors before final integration. Select a suitable measurement range for the expected pressure differences.
Pressure tubing or pressure taps cause incorrect measurements	Mid	High	Perform leakage and blockage checks, use consistent tubing and verify every pressure channel before testing.
Electrical or software integration problems delay testing	Mid	Mid	Develop and test the DAQ, sensor system and plotting software as separate modules before full integration.
Experimental Cp differs significantly from theory	Mid	Mid	Check flow uniformity, sensor calibration, pressure reference and model alignment. Repeat measurements and document remaining experimental limitations.
Components or sensors are unavailable within the project period			Reuse available hardware where suitable and define the minimum viable measurement setup early in the project.

Dependencies
The project depends on the available project time, budget and delivery times of required components. Long or unexpected delivery times for sensors, electronics or mechanical components may delay the implementation and testing phases. The available budget limits the selection of components. Where possible, existing hardware and laboratory equipment will therefore be reused to reduce both costs and lead times.

Activities by project phase	
Analysis	Inspect the complete existing wind tunnel and document the current mechanical, electrical and software condition.
Analysis	Define functional requirements and acceptance criteria based on successful measurement.
Research	Research pressure coefficient theory and practical Cp measurement methods.

Research	Research suitable pressure sensors and required measurement ranges/resolution.
Research	Investigate software methods for automatic Cp calculation and plotting.
Research	Investigate suitable DAQ hardware and communication methods.
Research	Investigate the existing fan performance and major pressure losses in the tunnel.
Research	Investigate airflow quality, turbulence and flow-conditioning methods applicable to the existing tunnel.
Design	Design the complete pressure-measurement and DAQ architecture.
Design	Design the mechanical integration of the test model, pressure taps, tubing and sensors.
Design	Design the data-processing and Cp -visualisation software.
Implementation	Repair and modify the required mechanical and airflow-related components of the tunnel.
Implementation	Install and integrate the pressure measurement, electronics and software systems.
Testing	Calibrate and verify all pressure measurement channels.
Testing	Measure airflow velocity and evaluate test-section flow conditions.
Testing	Perform initial pressure measurements on the test model.
Testing	Generate the experimental Cp curve.
Testing	Repeat the experiment to evaluate measurement repeatability.
Testing	Compare the experimental Cp distribution with XFOIL/theoretical results and analyse deviations.
Recommendations	